

Mathematical & Algorithmic Formulations

COMPREHENSIVE EQUATION SHEETS FOR NEURAL NETWORKS & AI GOVERNANCE

Section 1: Artificial Neural Network Foundations

1. LINEAR COMBINATION (PRE-ACTIVATION NODE INPUT)

Computes the weighted sum of all incoming feature inputs, injecting a scalar bias baseline to control the neuron activation threshold.

$$z = \sum_{i=1}^n w_i x_i + b = w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b$$

z	The net scalar intermediate input forwarded to the activation function.
w_i	The weight parameter vector assigned to the input connection, mapping feature importance.
x_i	The layer input scalar or feature value from the dataset matrix.
b	The bias variable, allowing the hyper-plane boundary to slide independently of raw inputs.

2. BINARY STEP (THRESHOLD ACTIVATION)

A rigid activation function providing discrete state boundaries, predominantly used in initial historical Perceptron schemas.

$$f(z) = \{ 1 \text{ if } z \geq 0 ; 0 \text{ if } z < 0 \}$$

3. SIGMOID ACTIVATION FUNCTION

Maps unconstrained real values into a crisp probability distribution scale between 0 and 1. Highly utilized in binary classification output configurations.

$$f(z) = 1 / (1 + e^{-z})$$

4. HYPERBOLIC TANGENT (TANH) ACTIVATION FUNCTION

A zero-centered alternative mapping mechanism, compressing continuous inputs to a signed spectrum between -1 and 1.

$$f(z) = (e^z - e^{-z}) / (e^z + e^{-z})$$

5. RECTIFIED LINEAR UNIT (RELU) & LEAKY RELU

The global industry standard hidden layer structural transformer. Leaky formats preserve a slight negative slope to combat dead neuron gridlocks.

$$\text{ReLU: } f(z) = \max(0, z)$$

$$\text{Leaky ReLU: } f(z) = \max(0.01z, z)$$

6. MEAN SQUARED ERROR (MSE COST FUNCTION)

Quantifies the quadratic distribution variance between continuous target variables and structural model evaluations across dataset matrices.

$$J(W, b) = 1 / (2n) \cdot \sum_{i=1}^n (y^{(i)} - \hat{y}^{(i)})^2$$

$J(W, b)$	Total loss scalar output calculated over the total sample batch size.
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$\hat{y}^{(i)}$	The model's explicit forward-pass prediction vector output.
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$y^{(i)}$	The targeted ground-truth true labeled training value.
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n	The absolute count size of the current evaluated training batch.
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7. BACKPROPAGATION CHAIN RULE EXPANSION

Traces global downstream total error changes sequentially back through intermediate layer elements using partial composite derivatives.

$$\partial J / \partial w_{ij} = (\partial J / \partial \hat{y}) \cdot (\partial \hat{y} / \partial z) \cdot (\partial z / \partial w_{ij})$$

8. PARAMETER OPTIMIZATION GRADIENT DESCENT STEP

Updates architectural parameter dimensions by stepping down computed error curves relative to localized feature slopes.

$$w_{new} = w_{old} - \alpha \cdot (\partial J / \partial w_{old})$$

α	The Learning Rate tuning hyperparameter, managing total optimization step adjustments.
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$\partial J / \partial w$	The evaluated localized partial derivative slope metric.
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Section 2: Responsible AI & Parametric Governance

9. PARAMETRIC INTERPRETABILITY LINEAR BASE

Explicit, intrinsically explainable parametric formulation where structural mechanics can be directly determined from variable coefficients.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_m x_m$$

10. DISPARATE IMPACT RATIO (SAGEMAKER CLARIFY EQUITY METRIC)

Used to flag algorithmic selection bias across demographics. Values below 0.8 identify systematic adverse treatment of unprivileged classes.

$$\textit{Disparate Impact} = P(\hat{Y} = 1 \mid \textit{Unprivileged}) / P(\hat{Y} = 1 \mid \textit{Privileged})$$

$P(\hat{Y} = 1)$

The computed evaluation rate of assigning a positive or beneficial class prediction outcome (e.g., loan approval).